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function ps_spatial_unwrapping_plots

% Function to plot spatial unwrapping networks
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% ------------------------------------------------------------------------
% File............: ps_spatial_unwrapping_plots.m
% Version & Date..: 1.7.2.16, 12-DEC-2009
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% ------------------------------------------------------------------------
%
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% intended for scientific use only. Applications for commercial use are
% prohibited.
%
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%
% Change log
% v1.7.2.8, Freek van Leijen
% - improvement of plots
% v1.7.2.8, Freek van Leijen
% - unwrap2_istep
%


global project_id results_id fig unwrap_istep unwrap2_istep

fid = fopen([project_id '_project.mat'],'r');
if fid>0
  fclose(fid);
  load([project_id '_project.mat'])

  % ------------------------------------------------------------------------
  % Read data
  % ------------------------------------------------------------------------

  delta_r = max(1,round(sqrt(Nlines*Npixels)/1000));
  delta_az = max(1,delta_r*round(r_spacing/az_spacing));
  Nlines_new = ceil(Nlines/delta_az);
  Npixels_new = ceil(Npixels/delta_r);

  mrm = repmat(uint8(NaN),Nlines_new,Npixels_new);

  Nlines_buffer = min(Nlines,floor(max_mem_buffer/(8*Npixels)));
  Nbuffer = ceil(Nlines/Nlines_buffer);
  Nlines_rem_buffer = rem(Nlines,Nlines_buffer);

  mrm_fid = fopen([project_id '_mrm_uint8.raw'],'r');
  sl = 1;
  el = 0;
  for v = 1:Nbuffer
    if (v==Nbuffer)&&(Nlines_rem_buffer~=0)
      Nlines_buffer = Nlines_rem_buffer;
    end
    temp = fread(mrm_fid,[Npixels Nlines_buffer],'*uint8')';
    temp = temp(1:delta_az:end,1:delta_r:end);
    Nlines_buffer_new = size(temp,1);
    el = el+Nlines_buffer_new;
    mrm(sl:el,:) = temp;
    sl = sl+Nlines_buffer_new;
  end
  fclose(mrm_fid);
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  switch orbit
    case 'desc'
      mrm = flipud(fliplr(mrm));
  end

  fullscreen = [1 1 1280 1024];
  %fullscreen=get(0,'Screensize');
  [minratio,ratio_index] = min([fullscreen(3)/Npixels_new
fullscreen(4)/Nlines_new]);
  if ratio_index(1)==1 %(1) to prevent multiple maxima
    figpos = [1 1 fullscreen(3) fullscreen(3)*Nlines_new/Npixels_new];
  elseif ratio_index(1)==2
    figpos = [1 1 fullscreen(4)*Npixels_new/Nlines_new fullscreen(4)];
  end

  for z = 1:Npsc_selections
    for w = 1:unwrap2_istep

      %%%%%%%%%%% network psc %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
      fid = fopen([project_id '_network_psc_sel' num2str(z) '_' results_id ...
               '_unwrap_istep' num2str(unwrap_istep) ...
               '_unwrap2_istep' num2str(w) '.mat'],'r');
      if fid>0
      fclose(fid);
      load([project_id '_network_psc_sel' num2str(z) '_' results_id ...
            '_unwrap_istep' num2str(unwrap_istep) ...
            '_unwrap2_istep' num2str(w) '.mat'])

      psc_az = psc_az_orig;
      psc_r = psc_r_orig;

      switch orbit
       case 'desc'
          psc_az = Nlines-psc_az+1;
          psc_r = Npixels-psc_r+1;
      end

      Narcs = size(dpsc_arcs,1);

      % ---------------------------------------------------------------------
      % Plot results
      % ---------------------------------------------------------------------

      fig = fig+1;
      figure(fig);hold on;
      set(gcf,'visible','off');

set(gcf,'position',figpos,'PaperPositionMode','auto','InvertHardcopy','off');
      subimage(mrm,gray(256));

      plot(psc_r/delta_r,psc_az/delta_az,'ob','linewidth',2);

      for v = 1:Narcs
        plot(psc_r(dpsc_arcs(v,:))/delta_r,psc_az(dpsc_arcs(v,:))/delta_az,'b')
      end
      axis off
      hold off

      print('-dpng',['plots/' project_id '_1_network_psc_sel' num2str(z) ...
                  '_' results_id '_unwrap_istep' num2str(unwrap_istep) ...
                  '_unwrap2_istep' num2str(w) '.png']);

      close(fig);
      end
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      %%%%%%%%%%%% apriori removed arcs %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
      fid = fopen([project_id '_apriori_removed_arcs_sel' num2str(z) '_' ...
               results_id '_unwrap_istep' num2str(unwrap_istep) ...
               '_unwrap2_istep' num2str(w) '.mat'],'r');
      if fid>0
      fclose(fid);
      load([project_id '_apriori_removed_arcs_sel' num2str(z) '_' ...
            results_id '_unwrap_istep' num2str(unwrap_istep) ...
               '_unwrap2_istep' num2str(w) '.mat'])

      switch orbit
       case 'desc'
          psc_az = Nlines-psc_az+1;
          psc_r = Npixels-psc_r+1;
      end

      % ------------------------------------------------------------------------
      % Plot results
      % ------------------------------------------------------------------------

      fig = fig+1;
      figure(fig);hold on;
      set(gcf,'visible','off');

set(gcf,'position',figpos,'PaperPositionMode','auto','InvertHardcopy','off');
      subimage(mrm,gray(256));

      plot(psc_r/delta_r,psc_az/delta_az,'ob','linewidth',2);

      Narcs_orig = size(dpsc_arcs_orig,1);
      for v = 1:Narcs_orig
        plot(psc_r(dpsc_arcs_orig(v,:))/delta_r,...
             psc_az(dpsc_arcs_orig(v,:))/delta_az,'b')
      end

      Narcs_removed1 = size(dpsc_arcs_removed1,1);
      if Narcs_removed1~=0
        for v = 1:Narcs_removed1
          h1 = plot(psc_r(dpsc_arcs_removed1(v,:))/delta_r,...
                  psc_az(dpsc_arcs_removed1(v,:))/delta_az,...
                  'color',[255 140 0]./255,'linewidth',2);
        end
      else
        h1 = [];
      end

      Narcs_removed2 = size(dpsc_arcs_removed2,1);
      if Narcs_removed2~=0
        for v = 1:Narcs_removed2
          h2 = plot(psc_r(dpsc_arcs_removed2(v,:))/delta_r,...
                  psc_az(dpsc_arcs_removed2(v,:))/delta_az,...
                  'r','linewidth',2);
        end
      else
        h2 = [];
      end

      if ~isempty(h1)&~isempty(h2)
        legend([h1 h2],[{'Apriori removed'},{'Untestable'}],...
             'Location','SouthOutside','Orientation','horizontal');
      elseif ~isempty(h1)
        legend([h1],[{'Apriori removed'}],...
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          'Location','SouthOutside','Orientation','horizontal');
      elseif ~isempty(h2)
        legend([h2],[{'Untestable'}],...
          'Location','SouthOutside','Orientation','horizontal');
      end

        axis off

      print('-dpng',['plots/' project_id '_2_apriori_removed_arcs_sel' ...
                  num2str(z) '_' results_id '_unwrap_istep' ...
                  num2str(unwrap_istep) ...
                  '_unwrap2_istep' num2str(w) '.png'])
      close(fig);
      end


      %%%%%%%%%%% removed arcs %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
      fid = fopen([project_id '_removed_arcs_sel' num2str(z) '_' results_id ...
               '_unwrap_istep' num2str(unwrap_istep) ...
               '_unwrap2_istep' num2str(w) '.mat'],'r');
      if fid>0
      fclose(fid);
      load([project_id '_removed_arcs_sel' num2str(z) '_' results_id ...
            '_unwrap_istep' num2str(unwrap_istep) ...
            '_unwrap2_istep' num2str(w) '.mat'])

      switch orbit
       case 'desc'
          psc_az = Nlines-psc_az+1;
          psc_r = Npixels-psc_r+1;
          psc_az_new = Nlines-psc_az_new+1;
          psc_r_new = Npixels-psc_r_new+1;
          ref_array(:,2) = Nlines-ref_array(:,2)+1;
          ref_array(:,3) = Npixels-ref_array(:,3)+1;
      end

      % ------------------------------------------------------------------------
      % Plot results
      % ------------------------------------------------------------------------

      fig = fig+1;
      figure(fig);hold on;
      set(gcf,'visible','off');

set(gcf,'position',figpos,'PaperPositionMode','auto','InvertHardcopy','off');
      subimage(mrm,gray(256));

      plot(psc_r_new/delta_r,psc_az_new/delta_az,'ob','linewidth',2);

      Narcs_new = size(dpsc_arcs,1);
      for v = 1:Narcs_new
        plot(psc_r(dpsc_arcs(v,:))/delta_r,psc_az(dpsc_arcs(v,:))/delta_az,'b')
      end

      Nref = size(ref_array,1);
      for v = 1:Nref
        h1 = plot(ref_array(v,3)/delta_r,ref_array(v,2)/delta_az,'g^', ...
                'markersize',10,'linewidth',2,'MarkerFaceColor','none');
      end

      Nadapt = size(adapt_vec,1);
      if Nadapt~=0
        for v = 1:Nadapt
          h2 = plot(psc_r(dpsc_arcs(adapt_vec(v,1),:))/delta_r, ...
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                psc_az(dpsc_arcs(adapt_vec(v,1),:))/delta_az,'r', ...
                'linewidth',2);
      end
    else
      h2 = [];
    end

    if ~isempty(h2)
      legend([h1 h2],[{'Reference PS'},{'Removed'}],...
          'Location','SouthOutside','Orientation','horizontal');
    else
      legend([h1],[{'Reference PS'}],...
          'Location','SouthOutside','Orientation','horizontal');
    end

    axis off

    print('-dpng',['plots/' project_id '_3_removed_arcs_sel' num2str(z) ...
                '_' results_id '_unwrap_istep' num2str(unwrap_istep) ...
                '_unwrap2_istep' num2str(w) '.png'])
    close(fig);

    end


    %%%%%%%%%%% adapted arcs %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
    fid = fopen([project_id '_adapted_arcs_sel' num2str(z) '_' results_id ...
            '_unwrap_istep' num2str(unwrap_istep) ...
            '_unwrap2_istep' num2str(w) '.mat'],'r');
    if fid>0
    fclose(fid);
    load([project_id '_adapted_arcs_sel' num2str(z) '_' results_id ...
          '_unwrap_istep' num2str(unwrap_istep) ...
            '_unwrap2_istep' num2str(w) '.mat'])

    switch orbit
     case 'desc'
        psc_az = Nlines-psc_az+1;
        psc_r = Npixels-psc_r+1;
        ref_array(:,2) = Nlines-ref_array(:,2)+1;
        ref_array(:,3) = Npixels-ref_array(:,3)+1;
    end

    psc = unique(dpsc_arcs_new);
    psc_az_new = psc_az(psc);
    psc_r_new = psc_r(psc);

    % --------------------------------------------------------------------
    % Plot results
    % --------------------------------------------------------------------

    fig = fig+1;
    figure(fig);hold on;
    set(gcf,'visible','off');

set(gcf,'position',figpos,'PaperPositionMode','auto','InvertHardcopy','off');
    subimage(mrm,gray(256));

    plot(psc_r_new/delta_r,psc_az_new/delta_az,'ob','linewidth',2);

    Narcs_new = size(dpsc_arcs_new,1);
    for v = 1:Narcs_new

plot(psc_r(dpsc_arcs_new(v,:))/delta_r,psc_az(dpsc_arcs_new(v,:))/delta_az,'b')
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      end

      Nref = size(ref_array,1);
      for v = 1:Nref
        h1 = plot(ref_array(v,3)/delta_r,ref_array(v,2)/delta_az,'g^', ...
                'markersize',10,'linewidth',2,'MarkerFaceColor','none');
      end

      if ~isempty(adapt_vec)
        Nel = histc(adapt_vec(:,1),0.5:1:Narcs_new-0.5);
        cmap = flipud(hot(max(Nel)+1));
        for v = 1:Narcs_new
          plot(psc_r(dpsc_arcs_new(v,:))/delta_r,...
             psc_az(dpsc_arcs_new(v,:))/delta_az,'color',cmap(Nel(v)+1,:),...
             'linewidth',2);
        end
        colormap(cmap);
        k = colorbar;
        set(get(k,'title'),'string','Nadapt');
        set(k,'Ytick',[1:ceil(max(Nel)/5):max(Nel)+1]);
        set(k,'Yticklabel',cellstr(num2str([1:ceil(max(Nel)/5):max(Nel)+1]'-
1)));
      end

      legend([h1],[{'Reference PS'}],...
             'Location','SouthOutside','Orientation','horizontal');

      axis off

      print('-dpng',['plots/' project_id '_4_adapted_arcs_sel' num2str(z) ...
                  '_' results_id '_unwrap_istep' num2str(unwrap_istep) ...
                  '_unwrap2_istep' num2str(w) '.png'])

      close(fig);

      %%%%%%%%%%% final network %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

      fig = fig+1;
      figure(fig);hold on;
      set(gcf,'visible','off');

set(gcf,'position',figpos,'PaperPositionMode','auto','InvertHardcopy','off');
      subimage(mrm,gray(256));

      plot(psc_r_new/delta_r,psc_az_new/delta_az,'ob','linewidth',2);

      Narcs_new = size(dpsc_arcs_new,1);
      for v = 1:Narcs_new

plot(psc_r(dpsc_arcs_new(v,:))/delta_r,psc_az(dpsc_arcs_new(v,:))/delta_az,'b')
      end

      Nref = size(ref_array,1);
      for v = 1:Nref
        h1 = plot(ref_array(v,3)/delta_r,ref_array(v,2)/delta_az,'g^', ...
                'markersize',10,'linewidth',2,'MarkerFaceColor','none');
      end

      legend([h1],[{'Reference PS'}],...
             'Location','SouthOutside','Orientation','horizontal');

      axis off

      print('-dpng',['plots/' project_id '_5_final_network_sel' num2str(z) ...
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                    '_' results_id '_unwrap_istep' num2str(unwrap_istep) ...
                    '_unwrap2_istep' num2str(w) '.png'])

      close(fig);

      end

    end
  end
end
fclose('all');
close all
```